3.2 Implicit Differentiation

Name _____ Period ____ Date _____

AP Calculus 3.2 · Student sheet

I can…

- find dy/dx by implicit differentiation.
- find the slope and the tangent line of an implicit curve at a point.
- use the Power Rule with rational exponents.

Vocabulary · fill in the missing word

Implicit function — a function y = f(x) defined by an _____ in x and y that is not solved for y

Implicit differentiation — differentiate both sides with respect to x, _____ y as a function of x, then solve for $\frac{dy}{dx}$

Power Rule, rational exponents — $\frac{d}{dx}\,x^{p/q}$ = _____

Part 1 · Two halves of a circle

$x^2 + y^2 = 25$ at the point (3, 4).

Method	dy/dx	At (3, 4)
$y = \sqrt{(25 - x^2)}$		
$-x/y$		

Part 2 · Notes and examples (fill in each note, then work the example)

Implicit differentiation: differentiate both sides with respect to x. Each time you differentiate a y-term, multiply by _____.

Example 1

Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$, and the slope at (3, −4).

Solve for dy/dx: collect the dy/dx terms on one side and _____.

Example 2

Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 7$, and the slope at (1, 2).

Example 3

Find the tangent line to $x^2 + 3xy + y^2 = -1$ at (1, −2).

Second derivative: differentiate dy/dx again, then replace dy/dx with _____.

Example 4

Find $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 25$.

Rational exponents: d/dx x^(p/q) = _____.

Example 5

Find each derivative.

a. $y = x^{2/3}$

b. $y = \sqrt[3]{(x^2 + 1)}$

Part 3 · Partner practice

Partner 1

Find $\frac{dy}{dx}$ at (2, 1) on $x^2 - 2xy + 3y^2 = 3$.

Partner 2

(AP® Practice) If $x^2 + y^3 = 9$, then $\frac{dy}{dx}$ at the point (1, 2) is

(A) $-\frac{1}{6}$ (B) $-\frac{1}{3}$ (C) $\frac{1}{6}$ (D) $-\frac{2}{3}$